

Workbook: Respiratory System

Week 3 workbook • BIO 004 Human Anatomy • Dr. Sharilyn Rennie • Solano Community College

A comprehensive workbook on the anatomy of the respiratory system: the upper and lower airways, the larynx, the bronchial tree, the alveoli, the pleurae, and the basic mechanics of breathing. Basic functions are included; gas exchange chemistry and complex control belong elsewhere.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

Overview and divisions • 2 questions

Q01 • DOK 1 • Recall

List the major functions of the respiratory system.

Q02 • DOK 1 • Recall

Distinguish the upper and lower respiratory tract by structures and the conducting vs respiratory zones.

The upper airway • 3 questions

Q03 • DOK 2 • Apply

Three things happen to inhaled air in the nasal cavity. What are they, and what features make each possible?

Q04 • DOK 1 • Recall

Name the paranasal sinuses and what they do.

Q05 • DOK 1 • Recall

Name the three regions of the pharynx and what passes through each.

The larynx • 3 questions**Q06 • DOK 1 • Recall**

Name the major cartilages of the larynx and what each does.

Q07 • DOK 1 • Recall

What are the vocal cords (true vocal folds), what are they made of, and what do they do?

Q08 • DOK 2 • Apply

How does the larynx protect the lower airway during swallowing?

The lower airway • 5 questions

Q09 • DOK 1 • Recall

Describe the trachea's structure, including the cartilage that supports it.

Q10 • DOK 1 • Recall

What lines the trachea, and what is the mucociliary escalator?

Q11 • DOK 1 • Recall

Trace the airway from the trachea to the alveolus, naming each level.

Q12 • DOK 2 • Apply

Why does an inhaled foreign object tend to lodge in the right main bronchus more often than the left?

Q13 • DOK 2 • Apply

Bronchioles do not have cartilage in their walls. What do they have instead, and why does it matter?

Alveoli and gas exchange • 4 questions

Q14 · DOK 1 · Recall

What is an alveolus, and what kind of epithelium lines it?

Q15 · DOK 1 · Recall

What are Type I and Type II alveolar cells, and what does each do?

Q16 · DOK 2 · Apply

What is the respiratory (blood-air) barrier, and what cell layers and structures must oxygen cross to reach the blood?

Q17 · DOK 3 · Reason

Premature infants often lack surfactant. Predict the clinical consequence and explain it using alveolar anatomy.

Lungs, pleurae, and breathing · 5 questions

How many lobes does each lung have, and what divides them?

Describe the pleurae of the lungs.

What happens during quiet inspiration in terms of muscle action and thoracic volume?

How does quiet expiration happen?

Air entering the pleural cavity through a chest wound causes the lung on that side to collapse. Using what you know about pleural anatomy, explain why.